QUESTION BANK

<table>
<tr><th>Qno</th><th>Question</th></tr>
<tr><td>1</td><td>Solve
a. In a class of 10 boys and 5 girls. A committee of 4 students is to be selected. Find the probability for the committee to contain at least 3 girls.
b. If a number 'x' is selected from the natural numbers 1, 2, 3, . . . 100. Find the probability of 'x' to follow:
$$x + \frac{100}{x} > 29$$</td></tr>
<tr><td>2</td><td>If $P(A) = \frac{3}{8}, P(\overline{B}) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$. Find
a . $P(A/B)$ and $P(B/A)$
b . $P(\bar{A}/\overline{B})$ and $P(\overline{B}/\bar{A})$</td></tr>
<tr><td>3</td><td>A Bolt factory Machines A, B & C manufacture 20%, 30% & 50% of the total of their output and 6%, 3% & 2% are defective. A bolt is drawn at random and found to be defective. Find the probabilities that it is manufactured from
Machine - A , Machine – B and Machine - C.</td></tr>
<tr><td>4</td><td>In a certain college, 25% of boys and 10% of girls are studying Mathematics. The girls constitute 60% of the student body.
a. What is the probability that mathematics is being studied?
b. If a student is selected at random and is found to be studying mathematics, Find the probability that the student is a girl?</td></tr>
<tr><td>5</td><td>A random variable X has the following probability function.
<table>
<tr><td>X=x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr>
<tr><td>P(X=x)</td><td>0</td><td>k</td><td>2k</td><td>2k</td><td>3k</td><td>k^2</td><td>$2k^2$</td><td>$7k^2+k$</td></tr>
</table>
a. Determine k and Evaluate P (X < 6)
b. Find Mean, Variance and also Find the cumulative distribution function.</td></tr>
<tr><td>6</td><td>If a random variable has the probability density function f(x) as:
$$f(x) = \begin{cases} 2e^{-2x} & for\ x > 0 \\ 0 & for\ x < 0 \end{cases}$$
Find the probability that it takes the values
a. (i). Between 1 and 3 and (ii) greater than 0.5
b. Find Mean and Find Variance.</td></tr>
<tr><td>7</td><td>Ten unbiased coins are tossed simultaneously. Obtain the probability of:
a. (i). Exactly 6 heads and (ii). At least 8 heads
b. (i). No Heads and (ii). More than 3 heads</td></tr>
</table>

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8	The mean of binomial distribution is 3 and the variance is $\frac{9}{4}$. Find a. (i). $P(X \geq 4)$ (ii). $P(1 \leq X < 6)$ b. (i). $P(X \leq 3)$ (ii). $P(4 < X \leq 7)$
9	If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals a. (i). exactly 3 individuals suffer from bad reaction (ii). more than 2 individuals suffer from bad reaction b. (i). None suffer from bad reaction (ii). More than one individual suffers from bad reaction
10	A car hire firm has two cars which it fires out day by day. The number of demands for a car on each day is distributed as Poisson variate with mean 1.5 calculate the proportion of days on which a. Neither car is used b. Some demand is refused.
11	Given a standard normal distribution, Find the area under the curve that lies a. (i). To the left of z = 1.43 (ii) To the right of z =-0.89 b. (i). Between z = -0.48 and z = 1.74 (ii). Between z = -2.16 and z = -0.65
12	Mean heights of students is 158 cm's with a standard deviation of 20. Find a. How many students' heights lie between 150 cm's 170 cm's in a class of 100 students. b. How many students' heights is more than 170 cm's in a class of 100 students.
13	The mean yield for one-acre plot is 662 kgs with a standard deviation of 32 kgs. Assuming normal distribution, how many one- acre plots in a batch of 1000 plots would we expect to have a yield of: a. (i). Over 700 kgs and (ii) Below 650 kgs b. Between 675 kgs and 725 kgs.
14	Given a standard normal distribution, Find the value of k such that a. (i). $P(Z > k) = 0.2946$ and (ii) $P(Z < k) = 0.0427$ b. (i). $P(k < Z < -0.18) = 0.4197$ and (ii). $P(-0.93 < Z < k) = 0.7235$
15	In a normal distribution, 7% of the items are under 35 and 89% are under 63. a. Determine the mean and variance of the distribution

	b. Find the Probability that an item selected at random lies between 45 and 56
16	The marks of SDF subject are distributed normally with mean and standard deviation are 75 and 10 respectively, 20 % of total students are failed in exam. Top 15% of total students got A grade, then Find a. What are the minimum marks to get pass b. What are the minimum marks to get grade A.
17	Answer the following: a. What is the effect on standard error, if a sample is taken from an infinite population of sample size is increased from 400 to 800. b. A random sample of size 100 is taken from an infinite population having the mean $\mu = 76$ and the variance $\sigma^2 = 256$. What is the probability that $\bar{x}$ will be between 75 and 78
18	The guaranteed average life of a certain type of electric bulbs is 1500 hrs. with a S.D of 120 hrs. It is decided to sample the output so as to ensure that 95% of bulbs do not fall short of the guaranteed average by more than 2%. What will be the minimum sample size?
19	Answer the following: a. What is the maximum error one can expect to make with probability 0.90 when using the mean of a random sample of size n = 64 to estimate the mean of population with $\sigma^2 = 2.56$ b. It is desired to estimate the mean number of hours of continuous use until a certain computer will first require repairs. If it can be assumed that σ = 48 hours, how large a sample be needed so that one will be able to assert with 90% confidence that the sample mean is off by at most 10 hours.
20	The pulse rate of 50 yoga practitioners decreased on the average by 20.2 beats/minute with S.D of 3.5. a. If $\bar{x} = 20.2$ is used as a point estimate of the true average decrease in pulse rate. What can we assert with 95% confidence about the maximum error E. b. Construct a 99% confidence interval for the true average decrease in pulse rate.
21	A sample of 900 members has a mean of 3.4 cm's and S.D 2.61 cm's. Is this sample has been taken from a large population of mean 3.5 cm's. Given that the population is normally distributed. .
22	The mean breaking strength of cables supplied by a manufacturer is 1800 with standard deviation 100. By a new technique in the manufacturing process, it is claimed that the breaking strength of the cable has increased. In order to test the claim a sample of 50 cables is tested. It is found that the mean breaking a strength is 1850. Can we support that claim

	at 1% level of significance.
23	A random sample of 100 electric bulbs, produced by a manufacturer A showed a mean life of 1190 hours with a S.D of 90. Another sample of 75, electric bulbs produced by a manufacturer B showed a mean life of 1230 with S.D of 120 hrs. Find whether there is a significant difference between the mean.
24	A random sample of 10 boys had the following I.Q.'s: 70, 120, 110, 101, 88, 83, 95, 98, 107 and 100. Do these data support the assumption of a population mean I.Q. of 100.
25	Two horses A and B were tested according to the time (in seconds) to run a particular track with the following results.

Horse-A	28	30	32	33	33	29	34
Horse-B	29	30	30	24	27	29	

Test whether the two horses have the same running capacity.

26	The average losses of workers, before and after certain program are given below. Use 0.05 level of significance to test whether the program is effective (paired sample t-test). 40 and 35, 70 and 65, 45 and 42, 120 and 116, 35 and 33, 55 and 50, 77 and 73.
27	The measurements of the output of two units have given the following results. Assuming that both samples have been obtained from the normal populations at 5% significant level, test whether the two populations have the same variance.

Unit-I	14.1	10.1	14.7	13.7	14.0
Unit-II	14.0	14.5	13.7	12.7	14.1

28	In one sample of 10 observations, the sum of the squares of the deviations of the sample values from sample mean was 120 and in the other sample of 12 observations, it was 314. Test whether the difference of variance is significant at 5% level?
29	A pair of dice are thrown 360 times and the frequency of each sum is indicated below:

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	8	24	35	37	44	65	51	42	26	14	14

Would you say that the dice are fair on the basis of the chi-square test at 0.05 level of significance?

30	From the following data, find whether there is any significant liking in the habit of taking soft drinks among the categories of employees.

Soft Drinks	**Employees**		
	Clerks	Teacher	Officers
Coke	10	15	65
Thums Up	15	30	65
Fanta	50	60	30

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31 Find the straight line that best fits the following data

x	0	1	2	3	4
y	1	1.8	3.3	4.5	6.3

32 Fit a second-degree polynomial to the following data by the method of least squares.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

33 Fit a curve of the form ae^{bx} to the following data.

x	0	1	2	3
y	1.05	2.10	3.85	8.30

34 Obtain a relation of the form ab^{x} for the following data by the method of least squares points given in the following table

x	2	3	4	5	6
y	8.3	15.4	33.1	65.2	127.4

35 Calculate the coefficient of correlation between age of cars and annual maintenance cost and comment:

Age of Cars (years)	2	4	6	7	8	10	12
Annual Maintenance Cost(Rupees)	1600	1500	1800	1900	1700	2100	2000

36 Obtain the rank correlation coefficient for the following data.

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

37 Calculate the regression equations of Y on X from the data given below, taking deviations from actual means of X and Y.

Price (Rs.)	10	12	13	12	16	15
Amount Demand	40	38	43	45	37	43

Estimate the likely demand when the price is Rs. 20/-

38 A panel of two judges A and B graded dramatic performance by independently awarding marks as follows:

Performance No.	1	2	3	4	5	6	7
Marks by A	36	32	34	31	32	32	34
Marks by B	35	33	31	30	34	32	36

The eighth performance, however, which judge B could not attend, got 38 marks by judge A. If judge B had also been present, how many marks would he be expected to have awarded to the eighth performance?

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a. State true or false with justification. If two lines of regression are $x + 3y - 5 = 0$ and $4x + 3y - 8 = 0$ then the correlation coefficient is -0.5.

b. It is given that the means of x and y are 5 and 10. If the line of regression of y on x is parallel to the line $9x + 40 = 20y$, estimate the value of y for x = 30.

40 Find Karl Pearson's coefficient of correlation from the following data:

Wages	100	101	102	102	100	99	97	98	96	95
Cost of Living	98	99	99	97	95	92	95	94	90	91

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41	A machine repairing shop gets on average 16 machines per day (of eight hours) for repair and the arrival pattern is Poisson. The shop owner has two applicants A and B for the job of repairman. Both A and B claim that the service times are exponentially distributed with means 20 and 15 minutes respectively. They demand salaries Rs. 500 and Rs. 600 per day respectively. The lost time costs Rs. 50/- per hour per machine. Assuming that the claims of the applicants are true, which one should be employed?
42	A toll gate is operated on a frequency where cars arrive according to a Poisson distribution with mean frequency of 1.2 cars per minute. The time of completing payment follows an exponential distribution with mean of 20 seconds. Find a. (i). Idle time of the counter (ii). Average no. of cars in the system b. (i). Average no. of cars in the queue (ii). Average time that a car spends in the system
43	Barber A takes 15 minutes to complete one haircut. Customers arrive in his shop at an average rate of one every 30 minutes. Barber B takes 25 minutes to complete one haircut and customers arrive at his shop at an average rate of one every 50 minutes. The arrival process is Poisson and the service times follow an exponential distribution. Answer the following Where would you expect a bigger queue? Where would you require more time waiting included to complete a haircut?
44	A bank plans to open a single server drive-in banking facility at a certain center. It is estimated that 20 customers will arrive each hour on average. If on average, it requires 2 minutes to process a customer's transaction, determine: a. (i). The proportion of time that the system will be idle (ii). On the average, how long a customer will have to wait before reaching the server b. The fraction of customers who will have to wait.
45	Assume that both arrival rate and service rate following Poisson distribution. The arrival rate and service rate are 25 and 35 customers/hour respectively, at a single window in RTC reservation counter. Find ρ, L_s L_q, W_s and W_q
46	Workers come to a tool store room to enquiry about the special tools (required by them) for a particular job. The average time between the arrivals is 60 seconds and the arrivals are assumed to be in Poisson distribution. The average service time is 40 seconds. Find a. Average queue length b. Average length of non-empty queue.
47	Consider a single server queuing system with Poisson input and exponential service time. Suppose the mean arrival rate is 3 calling units per hour with the expected service time as 0.25 hours and the maximum permissible number of calling units in the system is two. Obtain the steady state probability distribution of the number of calling units in the system and then calculate the expected number in the system.
48	A car park contains 5 cars. The arrival of cars is Poisson with a mean rate of one per hour. The length of time each car spends in the car park has negative exponential distribution with mean

	2 hours. How many cars are in the car park on average and what is the probability of a newly arriving customer finding the car park full and having to park his car elsewhere?
49	A one-person barber shop has six chairs to accommodate people waiting for haircut. Assume that customers who arrive when all the six chairs are full leave without entering the shop. Customers arrive at the average rate of 3 per hour and spend an average of 15 minutes for service. Find a. (i). The probability that a customer can get directly into the barber chair upon arrival (ii). Expected number of customers waiting for a haircut b. (i). Effective arrival rate (ii). The time a customer can expect to spend in the barber shop.
50	A fast-food restaurant has one drive-in window. It is estimated that cars arrive according to a Poisson distribution at the rate of 1 every 5 minutes and that there is enough space to accommodate a line of 10 cars. Other arriving cars can wait outside this space, if necessary. It takes 4 minutes on the average to fill an order, but the service time actually varies according to an exponential distribution. Determine the following: a. (i). The probability that the facility is idle (ii). The expected number of customers waiting to be served. b. (i). Effective arrival rate (ii). Effective service rate.